Radiology AI | Frontend architecture and engineering standards

Frontend specification package - October 2026 | Planning baseline, not a repository audit

Engineering architecture

Technology decisions - React + TypeScript + Vite; React Router; TanStack Query for server state; React Hook Form and Zod for validated forms; accessible UI primitives; charting and visualization libraries selected after prototype testing.

Code organization - Use feature-first modules: app, shared/ui, shared/api, shared/auth, shared/i18n, features/patients, features/reports, features/ai, features/worklists, tests. Separate domain DTOs from view models and keep fetch calls in typed API clients.

Backend contract - ASP.NET Core remains authoritative for permissions, workflows, report status, audit logs, clinical data and AI jobs. Consume OpenAPI-generated types or clients where appropriate; never reproduce server business rules in the frontend.

Shared libraries - Framework and AI-Framework are .NET/backend libraries. Do not import them directly into React. Reuse their capabilities through secured backend endpoints, versioned DTOs, and documented API contracts.

Security - OIDC/OAuth2 authorization-code with PKCE where supported; prefer secure server-side sessions/BFF when practical; short-lived credentials; role/tenant enforcement on backend; no PHI in browser logs, telemetry, localStorage, URLs or offline caches; explicit logout and idle handling.

Quality - TypeScript strict mode, ESLint, Prettier, unit/component tests with Vitest and React Testing Library, Playwright end-to-end tests, axe accessibility checks, CI build/typecheck/test and dependency scanning.

Repo blueprint

Proposed directories - src/app; src/features/patients; src/features/worklist; src/features/reports; src/features/ai; src/features/imaging; src/shared/api; src/shared/ui; src/shared/auth; src/shared/i18n; src/test; public.

Environments - Local mock, development API, staging and production configured through deployment environment; never ship secrets in Vite variables.

CI gates - Lint, format, typecheck, unit tests, E2E smoke tests, dependency scan and production build required before merge.